

# ICSR Processing and Case Narrative Portfolio

## PubMed-Sourced Cases

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This portfolio presents three Individual Case Safety Reports processed from real peer-reviewed PubMed publications. Each case is processed in full accordance with ICH E2A guidelines, demonstrating competency in ICSR lifecycle management, MedDRA coding, adverse event assessment, and case narrative writing.

**Standards Applied:** ICH E2A, ICH E6(R3), MedDRA Hierarchy, WHO-UMC Causality Categories

# CASE 1: Atorvastatin-Induced Rhabdomyolysis in a Patient with Liver Cirrhosis

**Source:** Bhattarai S, Pradhan SR, Bhattarai S. Low-dose atorvastatin therapy induced rhabdomyolysis in a liver cirrhosis patient: a case report. Ann Med Surg (Lond). 2023 Sep 1;85(10):5232-5234. PMID: 37811121

## Source Case Description

A 74-year-old male with liver cirrhosis secondary to alcohol use was prescribed Atorvastatin for hyperlipidemia. He developed muscle tenderness and decreased muscle power 2 weeks following initiation of statin therapy. Creatine phosphokinase (CPK) was measured at more than 22,000 IU/L. Urinalysis revealed blood positivity. Atorvastatin was ceased. Laboratory parameters normalized following drug discontinuation. No concurrent interacting medications were reported.

## Section 1: Case Validity

**Assessment: VALID** - All four ICH E2A minimum criteria met:

- Identifiable patient: 74-year-old male
- Identifiable reporter: Treating physician (medically qualified)
- Suspect drug: Atorvastatin
- Adverse event: Rhabdomyolysis

## Section 2: Patient and Drug Information

**Patient:** 74-year-old male

**Medical History:** Liver cirrhosis secondary to alcohol use

**Indication:** Hyperlipidemia

**Suspect Drug:** Atorvastatin (low dose, oral)

**Action Taken:** Atorvastatin permanently discontinued

**Outcome:** Recovered - laboratory parameters normalized after discontinuation

## Section 3: MedDRA Coding

**AE: Muscle tenderness**

PT: Myalgia, HLT: Musculoskeletal and connective tissue signs and symptoms, HLGT: Musculoskeletal and connective tissue disorders NEC, SOC: Musculoskeletal and connective tissue disorders

**AE: Decreased muscle power**

PT: Muscle weakness, HLT: Musculoskeletal and connective tissue signs and symptoms, HLGT: Musculoskeletal and connective tissue disorders NEC, SOC: Musculoskeletal and connective tissue disorders

**AE: Rhabdomyolysis**

PT: Rhabdomyolysis, HLT: Muscle related conditions NEC, HLGT: Musculoskeletal and connective tissue disorders NEC, SOC: Musculoskeletal and connective tissue disorders

**AE: CPK elevated**

PT: Blood creatine phosphokinase increased, HLT: Muscle and bone investigations, HLGT: Investigations NEC, SOC: Investigations

**AE: Blood in urine**

PT: Haematuria, HLT: Urinary tract signs and symptoms, HLGT: Urinary tract disorders, SOC: Renal and urinary disorders

## Section 4: Seriousness Assessment

**Assessment: SERIOUS** - Criteria met: Medically significant - CPK exceeding 22,000 IU/L indicating severe muscle breakdown with secondary haematuria, requiring immediate drug discontinuation. Rhabdomyolysis in the context of liver cirrhosis carries risk of acute kidney injury and is life-threatening.

## Section 5: Causality Assessment

**Assessment: PROBABLE (WHO-UMC)**

- Clear temporal relationship - onset 2 weeks following Atorvastatin initiation
- Positive dechallenge - laboratory parameters normalized after discontinuation
- Pre-existing liver cirrhosis impairs statin metabolism via reduced hepatic CYP3A4 activity, increasing systemic exposure and myotoxicity risk
- No interacting concomitant medications identified
- CPK greater than 10x ULN consistent with statin-induced rhabdomyolysis diagnostic criteria

## Section 6: Expectedness Assessment

**Assessment: EXPECTED (Labeled)** - Atorvastatin prescribing information lists myopathy and rhabdomyolysis as known serious adverse reactions, with hepatic impairment identified as a risk factor. Event is consistent with reference safety information.

## Section 7: Case Narrative

This is a published case report concerning a 74-year-old male patient with a known history of alcohol-related liver cirrhosis prescribed Atorvastatin for hyperlipidemia.

Approximately 2 weeks following initiation of Atorvastatin therapy, the patient developed muscle tenderness and decreased muscle power. Investigations revealed creatine phosphokinase levels exceeding 22,000 IU/L and blood-positive urinalysis, consistent with a diagnosis of rhabdomyolysis.

Atorvastatin was permanently discontinued. Laboratory parameters normalized following drug cessation. No rechallenge was performed.

This case is assessed as serious based on medical significance, with CPK exceeding 10x upper limit of normal and risk of secondary acute kidney injury. Causality is assessed as probable based on clear temporal association, positive dechallenge, and the established pharmacokinetic mechanism of impaired statin metabolism in hepatic dysfunction. The events are assessed as expected based on current Atorvastatin reference safety information.

# CASE 2: Low-Dose Methotrexate Toxicity Causing Bone Marrow Suppression

**Source:** Khuwaja S, Lyons M, Zulfiqar B. A Rare Case of Acute Methotrexate Toxicity Leading to Bone Marrow Suppression. Case Rep Med. 2024;2024:7693602. DOI: 10.1155/2024/7693602. PMID: 38510534

## Source Case Description

A patient with rheumatoid arthritis on weekly low-dose Methotrexate accidentally took the medication daily for 5 consecutive days instead of once weekly. The patient subsequently developed severe pancytopenia with WBC  $1.2 \times 10^9/L$ , hemoglobin 7.8 g/dL, and platelets  $18 \times 10^9/L$ , along with oral mucositis and fever. The patient required hospitalization, leucovorin rescue therapy, G-CSF, and blood transfusions. Recovery was achieved over several weeks.

## Section 1: Case Validity

**Assessment:** **VALID** - All four ICH E2A minimum criteria met:

- Identifiable patient: Adult patient with rheumatoid arthritis
- Identifiable reporter: Treating physician (medically qualified)
- Suspect drug: Methotrexate (accidental daily overdose)
- Adverse events: Pancytopenia, mucositis, fever

## Section 2: Patient and Drug Information

**Indication:** Rheumatoid arthritis

**Suspect Drug:** Methotrexate - weekly low dose (taken daily in error for 5 days)

**Nature of Event:** Medication error - accidental overdose

**Action Taken:** Methotrexate discontinued; leucovorin rescue, G-CSF, and blood transfusions administered

**Outcome:** Recovered after several weeks of supportive treatment

## Section 3: MedDRA Coding

**AE: Low white cell count**

PT: White blood cell count decreased, HLT: Leukopaenias, HLGT: White blood cell disorders, SOC: Blood and lymphatic system disorders

**AE: Low haemoglobin**

PT: Anaemia, HLT: Anaemias NEC, HLGT: Red blood cell disorders, SOC: Blood and lymphatic system disorders

**AE: Low platelets**

PT: Thrombocytopenia, HLT: Platelet disorders, HLGT: Platelet disorders, SOC: Blood and lymphatic system disorders

**AE: Pancytopenia**

PT: Pancytopenia, HLT: Bone marrow failures, HLGT: White blood cell disorders, SOC: Blood and lymphatic system disorders

**AE: Oral mucositis**

PT: Mucosal inflammation, HLT: Gastrointestinal signs and symptoms, HLGT: Gastrointestinal conditions NEC, SOC: Gastrointestinal disorders

**AE: Fever**

PT: Pyrexia, HLT: Body temperature conditions, HLGT: Body temperature conditions and temperature intolerance, SOC: General disorders and administration site conditions

**AE: Medication error**

PT: Accidental overdose, HLT: Medication errors, HLGT: Product issues, SOC: Product issues

#### Section 4: Seriousness Assessment

**Assessment: SERIOUS** - Criteria met: Hospitalization required; life-threatening - severe pancytopenia with platelets of  $18 \times 10^9/L$  representing critical bleeding risk, requiring leucovorin rescue, G-CSF support, and blood transfusions. Bone marrow suppression from methotrexate accumulation is a potentially fatal toxicity.

#### Section 5: Causality Assessment

**Assessment: CERTAIN (WHO-UMC)**

- Direct causal relationship established - accidental daily dosing for 5 consecutive days resulted in cumulative toxic drug exposure
- Positive dechallenge - recovery achieved following leucovorin rescue and drug discontinuation
- Well-established mechanism - methotrexate inhibits dihydrofolate reductase, impairing DNA synthesis in rapidly dividing cells including bone marrow precursors
- No confounding medications or alternative cause identified
- Reporter assessment confirms medication error as causative event

#### Section 6: Expectedness Assessment

**Assessment: EXPECTED (Labeled)** - Methotrexate prescribing information lists bone marrow suppression, pancytopenia, and mucositis as known serious adverse reactions with black box warnings. Event is consistent with reference safety information.

#### Section 7: Case Narrative

This is a published case report concerning an adult patient with rheumatoid arthritis on stable weekly low-dose Methotrexate therapy.

Due to a medication error, the patient accidentally took Methotrexate daily for 5 consecutive days instead of the prescribed once-weekly schedule, resulting in cumulative drug accumulation. The patient subsequently developed severe pancytopenia with WBC  $1.2 \times 10^9/L$ , hemoglobin 7.8 g/dL, and platelets  $18 \times 10^9/L$ , along with oral mucositis and fever, consistent with methotrexate-induced bone marrow toxicity.

The patient was hospitalized and treated with leucovorin rescue therapy, G-CSF, and blood transfusions. Recovery was achieved over several weeks. Methotrexate was discontinued.

This case is assessed as serious based on hospitalization requirement and life-threatening severity of pancytopenia. Causality is assessed as certain based on the confirmed medication error, cumulative toxic exposure, established pharmacological mechanism, and positive dechallenge with leucovorin rescue. The events are assessed as expected based on current Methotrexate reference safety information.

# CASE 3: Vancomycin-Induced DRESS Syndrome with Multi-Organ Involvement

**Source:** Mallick D, Kaushik N, Goyal L, Chandramohan D, Simhadri P, Singh P. A Rare Case of Vancomycin-Induced Drug Reaction With Eosinophilia and Systemic Symptoms (DRESS) Syndrome. Cureus. 2024 Nov 5;16(11):e73088. DOI: 10.7759/cureus.73088. PMID: 39640371

## Source Case Description

A patient was initiated on intravenous Vancomycin for treatment of a serious bacterial infection. Approximately 3 weeks after initiation, the patient developed a diffuse maculopapular rash covering more than 50% of body surface area, high-grade fever (39.2 degrees C), cervical lymphadenopathy, and peripheral eosinophilia. Laboratory workup revealed elevated ALT at 3x ULN and elevated creatinine indicating renal involvement. DRESS syndrome was diagnosed using the RegiSCAR scoring system. Vancomycin was immediately discontinued and systemic corticosteroids were initiated. Gradual improvement was observed over 4 to 6 weeks.

## Section 1: Case Validity

**Assessment: VALID** - All four ICH E2A minimum criteria met:

- Identifiable patient: Adult patient
- Identifiable reporter: Treating physician (medically qualified)
- Suspect drug: Vancomycin IV
- Adverse events: DRESS syndrome with rash, fever, eosinophilia, hepatic and renal involvement

## Section 2: Patient and Drug Information

**Indication:** Serious bacterial infection

**Suspect Drug:** Vancomycin IV

**Onset:** Approximately 3 weeks after initiation

**Action Taken:** Vancomycin immediately discontinued; systemic corticosteroids initiated

**Outcome:** Partially recovered - gradual improvement over 4 to 6 weeks

## Section 3: MedDRA Coding

**AE: Diffuse rash**

PT: Rash maculo-papular, HLT: Epidermal and dermal conditions, HLGT: Epidermal and dermal conditions, SOC: Skin and subcutaneous tissue disorders

**AE: High fever**

PT: Pyrexia, HLT: Body temperature conditions, HLGT: Body temperature conditions and temperature intolerance, SOC: General disorders and administration site conditions

**AE: Swollen lymph nodes**

PT: Lymphadenopathy, HLT: Lymphadenopathies, HLGT: Lymphatic system disorders, SOC: Blood and lymphatic system disorders

**AE: High eosinophils**

PT: Eosinophilia, HLT: Eosinophilic disorders, HLGT: White blood cell disorders, SOC: Blood and lymphatic system disorders

**AE: Elevated liver enzymes**

PT: Alanine aminotransferase increased, HLT: Liver function analyses, HLGT: Liver function analyses, SOC: Investigations

**AE: Renal impairment**

PT: Blood creatinine increased, HLT: Renal function analyses, HLGT: Renal function analyses, SOC: Investigations

**AE: DRESS syndrome**

PT: Drug reaction with eosinophilia and systemic symptoms, HLT: Hypersensitivity conditions, HLGT: Immune system disorders NEC, SOC: Immune system disorders

## Section 4: Seriousness Assessment

**Assessment: SERIOUS** - Criteria met: Life-threatening - DRESS syndrome is a severe systemic hypersensitivity reaction with reported mortality of up to 10%. Multi-organ involvement including hepatic and renal dysfunction, rash covering greater than 50% BSA, and systemic inflammation required hospitalization and immunosuppressive treatment.

## Section 5: Causality Assessment

**Assessment: PROBABLE (WHO-UMC)**

- Temporal relationship consistent with DRESS latency - onset approximately 3 weeks after initiation, characteristic of the 2 to 9 week latency reported for DRESS
- Positive dechallenge - gradual improvement following Vancomycin discontinuation and corticosteroid initiation
- DRESS diagnosis confirmed using validated RegiSCAR scoring system
- Vancomycin is a well-documented causative agent of DRESS syndrome
- Multi-organ involvement consistent with DRESS phenotype
- No alternative drug or causative factor identified

## Section 6: Expectedness Assessment

**Assessment: EXPECTED (Labeled)** - Vancomycin prescribing information lists severe hypersensitivity reactions including DRESS syndrome as known serious adverse reactions. Event is consistent with reference safety information.

## Section 7: Case Narrative

This is a published case report concerning an adult patient initiated on intravenous Vancomycin for treatment of a serious bacterial infection.

Approximately 3 weeks following initiation of Vancomycin therapy, the patient developed a diffuse maculopapular rash covering more than 50% of body surface area, high-grade fever of 39.2 degrees C, cervical lymphadenopathy, and peripheral eosinophilia. Laboratory investigations demonstrated elevated ALT at 3x upper limit of normal and elevated creatinine indicating hepatic and renal involvement. DRESS syndrome was diagnosed using the validated RegiSCAR scoring system.

Vancomycin was immediately discontinued and systemic corticosteroid therapy was initiated. Gradual clinical improvement was observed over 4 to 6 weeks. The case was reported by the treating physician.

This case is assessed as serious based on the life-threatening nature of DRESS syndrome with multi-organ involvement requiring hospitalization and immunosuppressive therapy. Causality is assessed as probable based on the characteristic latency period, positive dechallenge, confirmed DRESS diagnosis via RegiSCAR scoring, and established causality of Vancomycin in DRESS. The events are assessed as expected based on current Vancomycin reference safety information.

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All cases sourced from peer-reviewed PubMed publications and processed in accordance with ICH E2A guidelines. MedDRA coding based on current hierarchy conventions. Causality assessed using WHO-UMC categories. Prepared as a self-initiated learning exercise to demonstrate ICSR processing competency.